

Lab 4, Act 3

How much monitoring is enough?

NRES 746 – name: _____ group: _____

Picking up where we left off

In Act 1 you showed that a straight-line fit to log counts recovers the rate of decline, with a spread you could predict from a formula. In Act 2 you saw that a wrong model can be harmless for a mild decline, badly wrong for a severe one, and that you can sometimes catch it by checking the data against datasets simulated from the fitted model. Now we use all of that to answer the question the agency actually cares about: what does it take to run a monitoring program that reliably detects the decline of the toad?

The design problem

The true abundance in year 0 is $N_0 = 1000$ (assume it is known). Each animal present is detected independently with probability 0.02 per person-day of searching. A monitoring **bout** is defined by the number of observers P and the number of days D , and bouts are repeated every s years ($s = 1, 2, 3, 4$, or 5), starting in year 1, for as long as they fit within the 25-year window ($t = 1, 1 + s, 1 + 2s, \dots \leq 25$). Costs are \$200 per person-day, plus \$2,000 to mount each field expedition, regardless of its length.

Each bout produces one count, y_t . The analysis is the one from Act 1: regress $\log(y_t)$ on t with `lm()`. A decline is **detected** if the slope estimate is negative and its two-sided p -value is at most 0.05. The management goal is to have **at least a 75% chance of detecting a decline of 25% or more over 25 years**.

What is the cheapest design that meets the goal, and how sure are we?

Try Part 1 on paper first, before you open a laptop or start typing code. Resist the urge to jump straight to R or an AI assistant. Once your plan is finalized, AI tools, `?help`, and web searches are all fair game for Part 2.

Part 1: Predict before you simulate (~30 min)

1a. For a single animal, what is the probability of being detected during a bout with P observers searching for D days? (Assume every observer-day is an independent 0.02 chance.) Evaluate it for $P = 1$, $D = 3$, and give the expected count of a bout in year 0 and its variance. (This is where the index $N_0 = 60$ in Acts 1 and 2 came from.)

1b. Your formula depends on P and D only through one combined quantity. What is it? Is that also true of the *cost* of a bout? What does that imply for how many design dimensions we really have, and what might make 5 observers for 1 day differ from 1 observer for 5 days in the real world?

Let $k = P \times D$ be the number of person-days per bout.

1c. Precision. For a positive random variable Y with mean μ and variance v , the delta method gives $\text{Var}(\log Y) \approx v/\mu^2$ (Bolker Appendix A.7). Apply this to a bout count with N animals present and per-animal detection probability q , and find the SD of $\log(y_t)$ as a function of N and q . Then evaluate it for $k = 3$ and $N = 860$ (about the abundance at the midpoint of the 25 years).

1d. For m surveys spaced s years apart, $S_{tt} = s^2 m(m^2 - 1)/12$. Fill in the table, and use your Act 1 result $SD(\hat{b}) = \sigma/\sqrt{S_{tt}}$, where σ is now your answer to 1c, to get the SD of the slope estimate for $k = 3$. (A calculator is fine.)

Interval s	Surveys m	S_{tt}	SD of slope, $k = 3$
1	25	1300	
2	13	728	
3			
4			
5			

1e. Power. A decline is detected when the slope estimate is negative and larger in magnitude than about 1.96 of its SDs. Ignoring the tiny chance of a significant *increase*, that gives

$$\text{power} \approx \Phi\left(\frac{|b|}{SD(\hat{b})} - 1.96\right),$$

where Φ is the standard normal cdf and $b = \ln(0.75)/25 = -0.0115$ is the true rate of decline. Some values of Φ : $\Phi(0) = 0.50$, $\Phi(0.25) = 0.60$, $\Phi(0.5) = 0.69$, $\Phi(0.75) = 0.77$, $\Phi(1) = 0.84$, $\Phi(1.25) = 0.89$, $\Phi(1.5) = 0.93$, $\Phi(2) = 0.98$. Approximate the power for $k = 3$ with annual surveys ($s = 1$) and with surveys every other year ($s = 2$).

1f. Design A is $k = 3$, $s = 1$. Design B is $k = 6$, $s = 2$. Which do you predict has higher power? (Hint: when q is small, how does the SD from 1c change when k doubles? How does S_{tt} change when s doubles?) Compute the total cost of each design. What does this suggest about how to spend a monitoring budget?

1g. Suppose the population is actually *stable*. How often will this analysis declare a decline? It is not 5%. Explain, and predict the number.

1h. In plain English (no R code), write the procedure you would follow to estimate the power of one design (k and s) by simulation, and the extra step you need to find the cheapest design that meets the goal.

Checkpoint: finalize your plan. Reread your procedure from 1h and your predictions from 1e-1g. Would your plan work if you started typing it up right now? Revise anything incomplete or unclear before you write code.

Part 2: Run the simulations (~60 min)

Use these settings: $N_0 = 1000$, per-animal detection probability 0.02 per person-day, $\lambda = 0.75^{1/25}$ so that the expected abundance is $N_t = \text{round}(N_0 \lambda^t)$, and analysis by `lm()` of $\log(y_t)$ on t . Detection is binomial: each bout count is a Binomial draw with size N_t and detection probability $1 - 0.98^k$.

2a. Write functions that (i) simulate the counts from one monitoring program, given k and s ; (ii) analyze them and report whether a decline was detected; and (iii) estimate the power of a design from R simulated programs. Record the prompt you used if you asked an AI for help, but not the AI's code.

2b. *Validate before you trust.* Set $\lambda = 1$ (no decline) and run $R = 2,000$ programs of the design $k = 3$, $s = 1$. Then compare your simulated power for $k = 3$ with $s = 1$ and $s = 2$ against your approximations in 1e.

Null-simulation detection rate: _____ $\pm$ _____ (Monte Carlo SE)

My prediction in 1g: _____

Simulated power, $s = 1$: _____ Approximation from 1e: _____

Simulated power, $s = 2$: _____ Approximation from 1e: _____

Does the null rate agree with your prediction? The approximations in 1e should be somewhat off. In which direction, and why? (Think about what the approximation ignored.)

2c. *The design space.* Estimate the power for $k = 1, \dots, 15$ and $s = 1, \dots, 5$ (1,000 programs per design is enough). Plot power against k , with one line for each s and a horizontal line at 0.75. Describe the pattern. Which of your predictions in 1f held up?

2d. Cost. Compute the cost of every design in your grid. Among designs with estimated power at or above 0.75, find the cheapest few. Because each power estimate has Monte Carlo error, re-estimate your top candidates with 5,000 programs each.

k	Interval s	Cost (\$)	Power (5,000 runs)	MC SE

Which design do you recommend, and why that one and not the cheapest one that squeaked past 0.75?

2e. Does the analysis scale matter here? Rerun your recommended design, but analyze the raw counts, y_t , with `lm()` instead of the log counts. Does the power change noticeably? Connect your answer to what you saw in Act 2 (hint: which scenario is this closer to, mild or severe?).

Power with raw counts: _____ Power with log counts: _____

2f. Optional extension: environmental noise. So far the only randomness in the abundance trajectory is sampling (detection) error. Real populations also experience year-to-year shocks that *persist*, such as a drought year that reduces the population, which the next year starts from. Modify your simulation so that $N_t = N_{t-1} \lambda e^{\varepsilon_t}$ with $\varepsilon_t \sim \text{Normal}(0, 0.05)$, and then rerun (i) the power of your recommended design and (ii) the detection rate when $\lambda = 1$. What happened, and which assumption from Act 1 is no longer true? What does this suggest about the kind of statistical model you would want for real monitoring data?

Power with environmental noise: _____ Detection rate when $\lambda = 1$: _____

Reflection (~10 min)

R1. Summarize your recommended design, its cost, and its estimated power. Which assumption of the simulation would you worry about most, and how would you find out whether it matters?

R2. Look back at the scenario in Act 1. Which result or derivation from Act 1 did you rely on most in this Act, and how?

R3. In three or four sentences, explain your recommendation to the agency's director, who has no statistical training. Then say what result would make *you* doubt your own recommendation.

R4. If you used an AI tool, how did you use it, and what did you have to check or correct? If you did not, what was hardest?
